

ANNEXURE - I
Brief Description of the Design
1D Convolutional Spiking Neural Network (CSNN) Architecture

Introduction

Convolutional Spiking Neural Networks (CSNNs) combine the event driven efficiency of Spiking Neural Networks (SNNs) with the feature extraction capabilities of convolutional architectures. Unlike traditional frame based Convolutional Neural Networks (CNNs), which process entire input tensors synchronously, the presented CSNN operates in an event driven, timestamp based manner. It receives serial spike events, applies a 1D convolutional kernel across input channels, and updates parallel Leaky Integrate and Fire (LIF) neurons. This hybrid enables low latency, power efficient temporal feature extraction, making it ideal for edge AI, gesture recognition, and real-time sensor processing.

The design implements a complete 1D convolutional SNN layer with:

- 16 input channels
- Kernel size of 3
- 16 output channels (parallel LIF neurons)
- 8 bit synaptic weights
- 16 bit membrane potential resolution

All output neurons operate in parallel, sharing the same convolutional weights and input spike window, while maintaining independent membrane potentials.

Workflow

The CSNN is a digital hardware realization of a 1D convolutional spiking layer. It integrates spike reception, temporal shift register buffering, parallel accumulate operations, and LIF neuron dynamics.

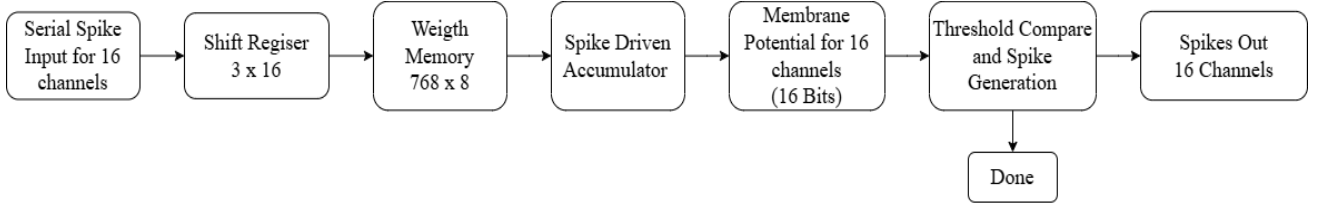


Figure: Block diagram of the 1D CSNN with parallel LIF neurons (16 input channels, 16 output channels, kernel size 3).

1. Spike Reception & Shift Register

A serial spike input is received over 16 consecutive clock cycles, building an 16 bit input vector. Upon completing a full input vector, the vector is shifted into a 3 stage deep shift register. This register stores the current and two previous input vectors, forming a 1D convolutional window of size 3×16 .

2. Parallel Weighted Accumulation

Once a full 3×16 window is available, the module enters the compute phase. For each of the 16 output channels independently:

- All $3 \times 16 = 48$ weights are conditionally accumulated based on the presence of a spike.
- Accumulation is performed in parallel across output channels using a combinational accumulator array, avoiding sequential bottlenecks.

3. LIF Neuron Update & Thresholding

After accumulation, each output channel updates its independent membrane potential:

- If membrane potential + accumulated weights $\geq$ threshold $\rightarrow$ spike output = 1 and membrane resets to 0.
- Otherwise $\rightarrow$ spike output remains 0 and membrane potential is incremented by accumulated weights

4. Configuration Interface

A synchronous configuration interface allows runtime programming:

- Writing to weights [0..767] ($16 \times 3 \times 16 = 768$ weights).
- Writing the global threshold value (default = 50).

Configuration is prioritised over spike reception to avoid contention.

5. Output & Handshake

After computation, the done signal is asserted for one clock cycle, and the 16 bit spikes_out vector is valid. The module returns to the receive state, ready for the next timestamp.

Table: Technical Specifications for 1D CSNN architecture

Technology node	180nm SCL PDK
Total Pins	62
Metal Layers	4
Power Supply	1.8V
Clock Frequency	20 MHz
Input Data Width	1 bit for spike in 10 bit weight address 8 bit for weight data
Output Data Width	16 bits for spike out
Size of chip	2mm x 2mm

Application

This 1D CSNN core is a fundamental building block for low power, event driven spiking neural networks. Typical use cases include:

- **PCG signal classification:** real time detection of normal vs. abnormal heart sounds (e.g., murmurs, valvular disorders) by converting phonocardiogram signals into spike trains and applying temporal convolution for feature extraction.
- Gesture recognition from event based sensors (e.g., DVS) the 1D convolution processes temporal slices of spike events.
- **Keyword spotting:** audio spike streams converted from analog signals; the temporal kernel captures phoneme level features.
- **Edge AI in wearables & IoT:** battery powered devices requiring continuous sensing without cloud offload.

The design is particularly suited for time-series spike classification where input arrives as a stream of binary events and latency is critical.